Bootstrap (L5 Supplementary Materials)

MFE5340: AI in FE: Quantitative Investment

Chuan Shi
School of Data Science

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香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



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Bootstrap: Origins and Concept

The bootstrap is a computer-based method for assigning measures of accuracy to statistical estimates (Efron and Tibshirani 1993).

- Introduced by Bradley Efron in 1979, the Bootstrap method revolutionized statistical inference.
- It estimates the variation of a statistic by **resampling from the data itself,** avoiding assumptions about the population distribution.
- With the power of modern computing, implementing Bootstrap has become straightforward.
- You have already seen this: We use bootstrap in Barra's eigenfactor adjustment!

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What Does "Bootstrap" Mean?

The term "Bootstrap" is derived from the idiom "pull yourself up by your bootstraps", meaning to improve one's situation through one's own efforts.

- In statistical estimation, Bootstrap means using only the available sample data, with no assumptions about the population's distribution (e.g., no normality assumption).
- The goal is to estimate the error of a statistic derived from the sample data.
- The central idea is that it may sometimes be better to draw conclusions about the population characteristics strictly from the sample, rather than relying on assumptions about the population (Mooney and Duval 1993).

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Key Technique: Resampling with Replacement

- The main technique is **resampling with replacement**, where we repeatedly sample from the data. (Why is "with replacement" important? We will address this shortly!)
- For example, if a sample contains 10 elements, after drawing one element, we place it back, making it possible to select the same element again.
- This is in contrast to "sampling without replacement", where once an item is drawn, it cannot be selected again.

```python
import numpy as np

# original sample
original_sample = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

# generate 5 bootstrap samples
n_samples = 5
bootstrap_samples = [np.random.choice(original_sample,
    size=len(original_sample), replace=True) for _ in range(n_samples)]

# print
for i, sample in enumerate(bootstrap_samples, 1):
    print(f"Bootstrap Sample {i}: {' '.join(map(str, sample))}")
```

```
Bootstrap Sample 1: 7 1 9 5 9 2 3 8 8 1
Bootstrap Sample 2: 9 9 3 6 8 3 9 9 5 2
Bootstrap Sample 3: 10 10 3 1 10 1 6 7 1 5
Bootstrap Sample 4: 4 8 2 5 8 9 10 1 10 8
Bootstrap Sample 5: 3 7 6 8 1 6 1 4 7 10
```


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- Suppose we have n dat points $x_1, x_2, \cdots, x_n$ drawn from a distribution F.

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The Empirical Bootstrap

- Suppose we have n dat points $x_1, x_2, \cdots, x_n$ drawn from a distribution F.
- An **empirical bootstrap sample** is a resample of the **same size**: $x_1^\star, x_2^\star, \cdots, x_n^\star$.

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- Think of the latter as a sample of size n drawn from the empirical distribution $F^\star$.

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- Think of the latter as a sample of size n drawn from the empirical distribution $F^\star$.
- For any statistic v (e.g. mean or median) computed from the original sample data, we can define a statistic $v^\star$ by the same formula but computed instead using the resampled data.

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The Bootstrap Principle

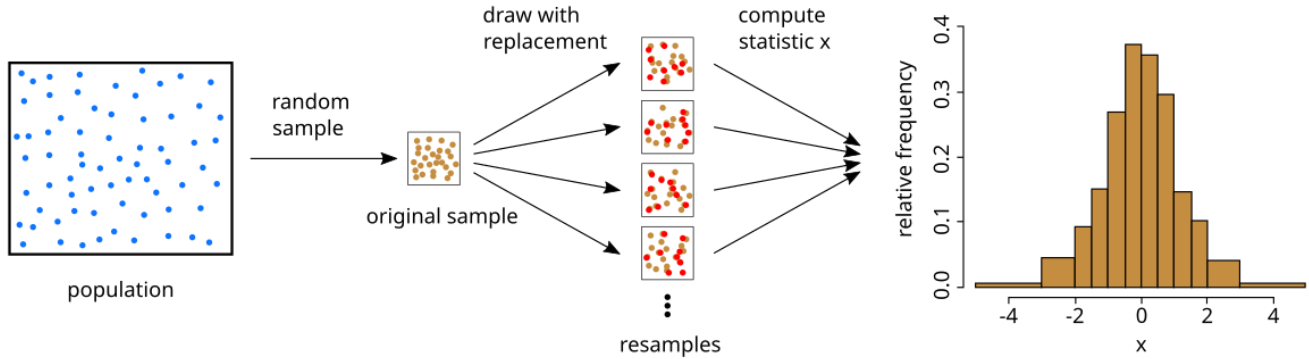
- $F^\star$ is approximately equal to F.
- The statistic v is well approximated by $v^\star$.
- The **variation of v around the true but unknown statistic u** of the population is well approximated by the **variation of $v^\star$ around v**.

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- $F^\star$ is approximately equal to F.
- The statistic v is well approximated by $v^\star$.
- The **variation of v around the true but unknown statistic u** of the population is well approximated by the **variation of $v^\star$ around v**.
- **Question**: Why is the resample the same size as the original sample? **Answer**: The variation of the statistic v depends on the size of the sample. If we want to approximate this variation, we need to use resamples of the same size. This also implies that we must resample with replacement. (Why?)

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The Bootstrap Principle



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Empirical Bootstrap Confidence Interval

- **Example adopted from MIT 18.05.**
- Goal: Estimate the mean μ of the underlying distribution and give an 80% bootstrap confidence interval.
- Sample data ($n = 10$): $\{30, 37, 36, 43, 42, 43, 43, 46, 41, 42\}$.
- Approach: Use empirical bootstrap to compute the confidence interval.

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- Use the sample mean $\bar{x} = 40.3$ as an estimate of μ. (This is the easy part.)
- To derive the confidence interval, we need to know how much the distribution of $\bar{x}$ varies around true mean μ: $\delta = \bar{x} - \mu$. (This is the hard part.)
- If we knew this distribution, we could find $\delta_{.1}$ and $\delta_{.9}$, the 0.1 and 0.9 critical values of δ. Then, we'd have:

$$\text{prob}(\delta_{.1} \leq \bar{x} - \mu \leq \delta_{.9}|\mu) = 0.8 \Leftrightarrow \text{prob}(\bar{x} - \delta_{.1} \geq \mu \geq \bar{x} - \delta_{.9}|\mu) = 0.8,$$

which gives an 80% confidence interval of

$$[\bar{x} - \delta_{.9}, \bar{x} - \delta_{.1}].$$

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- **BUT!** We don't know the distribution of δ.
- The bootstrap principle offers a practical approach to estimating the distribution of δ. It says we can approximate it by the distribution of:

$$\delta^\star = \bar{x}^\star - \bar{x},$$

where $\bar{x}^\star$ is the mean of a bootstrap sample.

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where $\bar{x}^\star$ is the mean of a bootstrap sample.

Since $\delta^\star$ is computed by resampling the original data, we can have simulate $\delta^\star$ as many times as we'd like. Then, we use the distribution of $\delta^\star$ to approximate the distribution of δ.

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Bootstrap Samples

- Generate 20 bootstrap samples, each of size 10.
- Each column represents one bootstrap sample:

43	36	46	30	43	43	43	37	42	42	43	37	36	42	43	43	42	43	42	43
43	41	37	37	43	43	46	36	41	43	43	42	41	43	46	36	43	43	43	42
42	43	37	43	46	37	36	41	36	43	41	36	37	30	46	46	42	36	36	43
37	42	43	41	41	42	36	42	42	43	42	43	41	43	36	43	43	41	42	46
42	36	43	43	42	37	42	42	42	46	30	43	36	43	43	42	37	36	42	30
36	36	42	42	36	36	43	41	30	42	37	43	41	41	43	43	42	46	43	37
43	37	41	43	41	42	43	46	46	36	43	42	43	30	41	46	43	46	30	43
41	42	30	42	37	43	43	42	43	43	46	43	30	42	30	42	30	43	43	42
46	42	42	43	41	42	30	37	30	42	43	42	43	37	37	37	42	43	43	46
42	43	43	41	42	36	43	30	37	43	42	43	41	36	37	41	43	42	43	43

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Compute $\delta^\star$

- Compute $\delta^\star = \bar{x}^\star - \bar{x}$ for each bootstrap sample.
- Sort $\delta^\star$ values from smallest to largest:

$$-1.6, -1.4, -1.4, -0.9, -0.5, -0.2, -0.1, 0.1, 0.2, 0.2, 0.4, 0.4, 0.7, 0.9, 1.1, 1.2, 1.2, 1.6, 1.6, 2.0$$

- Approximate the critical values:

$$\delta^\star_{0.1} = -1.4, \quad \delta^\star_{0.9} = 1.6.$$

- Compute the 80% confidence interval for μ:

$$[\bar{x} - \delta^\star_{.9}, \bar{x} - \delta^\star_{.1}] = [40.3 - 1.6, 40.3 + 1.4] = [38.7, 41.7].$$

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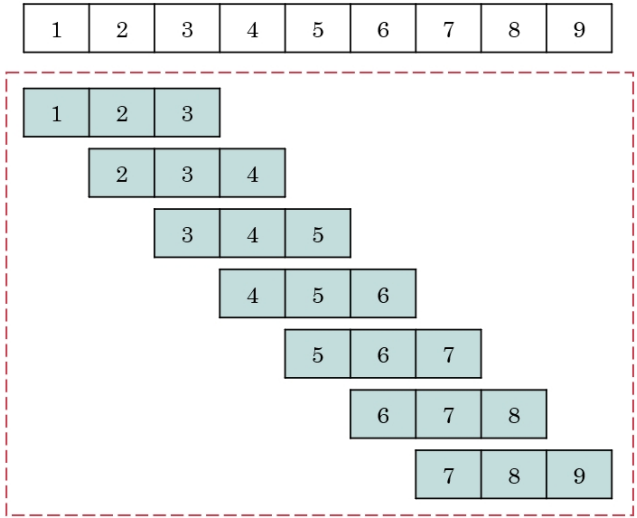
- Financial time series data (e.g. factor returns) may exhibit autocorrelation.
- Block Bootstrap is used to preserve autocorrelation during resampling.
- Popular Block Bootstrap methods:
  - Moving Block Bootstrap
  - Circular Block Bootstrap
  - Stationary Bootstrap

Block Bootstrap: Resample consecutive blocks of data to preserve autocorrelation.

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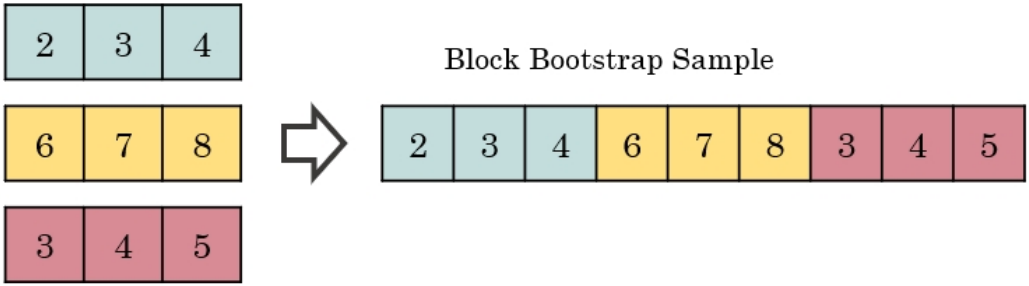
Moving Block Bootstrap

- Introduced by Kunsch (1989), Liu and Singh (1992).
- Blocks are formed by consecutive n data points from the time series.
- Example:
  - Original data: $1, 2, 3, 4, 5, 6, 7, 8, 9$
  - Block size: 3
  - Overlapping blocks: $\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 5\}, \ldots$



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- Resample blocks with replacement to generate a new bootstrap sample.
- First and last blocks may be under-sampled.



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Circular Block Bootstrap

- Introduced by Politis and Romano (1992).
- The key idea: Connect the first and last data points to form a circular structure.
- This avoids the problem of under-sampling at the edges of the data.
- Circular resampling is performed on the blocks, preserving the structure of the data.

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Stationary Bootstrap

- Introduced by Politis and Romano (1994).
- Unlike moving block bootstrap and circular block bootstrap, stationary bootstrap uses **non-fixed block sizes**.
- Block sizes follow a geometric distribution, with the expected block size given as input.
- Advantages: Better at handling non-stationary data and preserving the stationary structure of the data.

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Comparison of Block Bootstrap Methods

- Moving block bootstrap:
  - Fixed block size.
  - May cause under-sampling at edges.
- Circular block bootstrap:
  - Uses a circular structure.
  - Solves the edge under-sampling issue.
- Stationary bootstrap:
  - Non-fixed block size (geometrically distributed).
  - Works well with non-stationary data.

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Why Block Bootstrap for Financial Return Data?

- Autocorrelation in time series data:
  - Financial return data often exhibit autocorrelation, meaning that the returns at time t depend on returns at time $t-1$ (and beyond).
  - Standard resampling methods (e.g., simple random sampling) would destroy the temporal dependencies.
  - Block Bootstrap is used to preserve these dependencies by resampling blocks of consecutive observations.

- Besides, cross-sectional dependence:
  - Financial returns are also affected by cross-sectional correlations, such as between different assets or factors (e.g., value, momentum, or volatility).
  - Resampling multiple assets or factors together allows us to preserve cross-sectional dependence in addition to autocorrelation.
  - This is important because, in portfolio or factor modeling, returns are not independent across assets or factors.

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Questions?

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